

# CONFIGURING MULTI-AREA OSPF

## 2025



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Period 0 - 2

## Lab purpose:

By configuring multi area OSPF, we are learning a way to use OSPF in a way that is more efficient for larger networks. In addition, this lab has refamiliarized me with how networks work, and the role of routers to route packets between these networks

## Background:

Before OSPF, the main protocol in use was Routing Information Protocol (RIP). It had quickly become the standard after being included with the Berkeley standard Distribution (BSD) of UNIX as of 1982. However, it had many technical limitations, as the algorithm it used was a distance vector protocol, which ran on the bellman ford algorithm. The algorithm had many technical challenges, like no loop detection, long convergence time, as well as 16 being the value for infinity, that made RIP impossible to use in a situation where more than 15 hops were necessary in a network. As a result Open Shortest Path First, or OSPF was developed.

Ospf used Dijkstra's algorithm to find the shortest path between routers, and it allowed routes to be set dynamically based on the state of the network. OSPF is also an open standard, allowing anybody to implement it on their routers. This gives it a huge advantage over cisco proprietary protocols like EIGRP, which can only run on cisco hardware. This means that if you have other hardware, or need to interconnect your devices to non-cisco devices, you should be able to use OSPF most of the time, as it is widely supported among different brands of devices. This program is most useful in medium to smaller networks, as it has enough flexibility to allow network administrators to customize it to their needs. However, it falls short in extremely large networks, like the internet, as it does not have the tools like BGP does in order to handle the massive complexities of the internet.

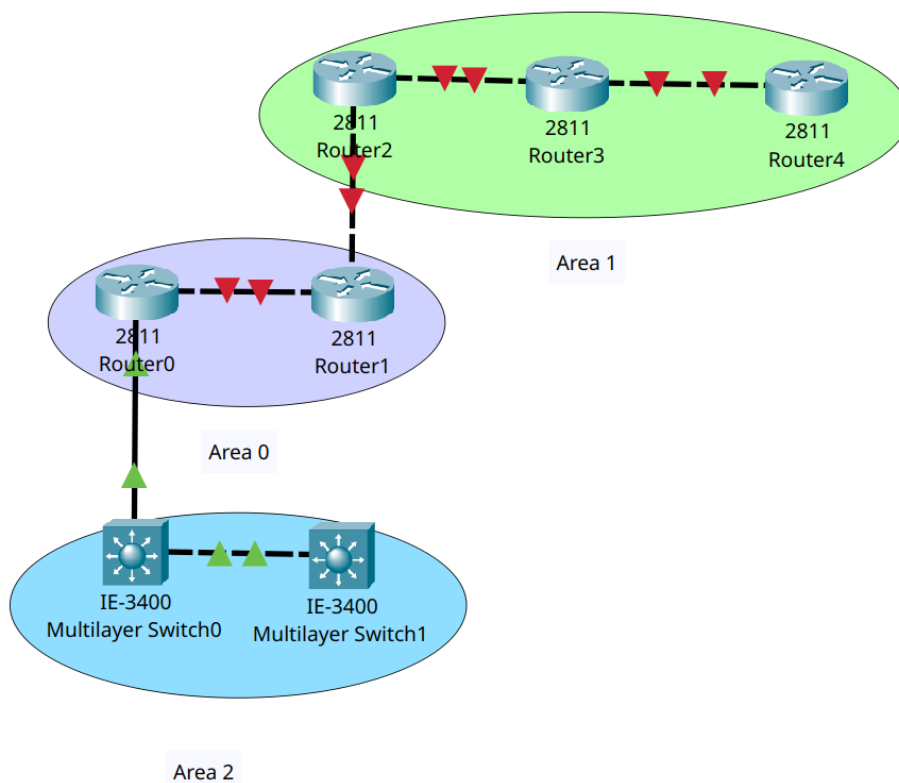
The design of OSPF requires each router to hold the entire network topology in order to have each router calculate the shortest path. This leads to a scalability issue as the routing information that each router needs to hold will also grow and potentially become unsustainable. In addition, if a link is flapping, it can lead to ripple effects over the entire OSPF network. To address these problems, multi-area OSPF was developed. Multi-area OSPF allows the OSPF network to be separated into multiple areas, with a mandatory backbone area 0. Each of the outer networks must be connected to area border routers (ABR) that are in area 0 to guarantee connectivity. Doing this allows each router to only be required to hold the network topology of its area in memory, reducing the memory and processing strain required to run OSPF. In addition, separating the network into separate areas also reduces the effects of flapping links to that specific area.

Overall, this hierarchical design leads to lower CPU load, lower RAM load, and faster convergence, as the routers are not experiencing the full load of potentially hundreds of routers, and tens of thousands of networks to calculate.

## Lab Summary:

In this lab, I configured multi area ospf in three areas, consisting of 5 routers and two multi layer switches.

## Lab topology:



## Lab commands:

```
ipv6 ospf <1-65535>
```

Starts ipv6 ospf processes on the router with the specified process id  
router-id A.B.C.D

Assigns Router-Id to ipv6 ospf process

```
router ospf <1-65535>
```

Starts ip ospf processes on the router with the specified process id  
router-id A.B.C.D

Assigns Router-Id to ip ospf process

```
network A.B.C.D area <area number>
```

Adds a network to ip ospf processing, and assigns it the specified area number

```
ipv6 unicast-routing
```

Enables ipv6 unicast-routing on Routers and Multilayer switches

```
ip routing
```

Enables ip routing on Routers and Multilayer switches

```
no switchport
```

Disables layer two behaviors on a layer three switches port

```
ipv6 ospf <1-65535> area <0-4294967295>
```

Enables ipv6 ospf processes on the specified area for the current interface

## Lab configurations:

R1 :

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
  exit-address-family
  address-family ipv6
  exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-702538597
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-702538597
  revocation-check none
  rsakeypair TP-self-signed-702538597
crypto pki certificate chain TP-self-signed-702538597
  certificate self-signed 01
    3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
    30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
    69666963 6174652D 37303235 33383539 37301E17 0D323530 39313932
32343033
    345A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403
1325494F
    532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3730
32353338
    35393730 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02
    82010100 D38CF1D2 172412CE 6489BD1A 3DC1726B 03509FA6 205C8655
F75E7C45

    E1C82059 575A871A 9695DC45 731664A7 829E451E B23DCBC7 D2707761
8BF8F888
    2648CF48 6F4AAB3F A091782D B4EB8819 36FFAE74 180F3495 5A624617
A60E943D
```

AFC943B7 8A6B03EF 81D86095 045D0166 DC68376E 91EB7E5B 34212588  
D20C8CCB  
37973C41 C3BF62CF F137A6AD BCFFC53F BC4C2EA0 654ECE5F 5EB85843  
CA11E792  
CD5DEBDE D4B2FAD2 8A09D58A 0FF50C90 BB90F0C8 B2199248 075BCBE3  
CBC71213  
48ED7D30 60E9FC72 B42C78A7 BD1F1269 248AA8C7 E466D3EC 80BF504C  
EB076FFD  
CAE78F99 02030100 01A35330 51300F06 03551D13 0101FF04 05300301  
01FF301F  
0603551D 23041830 16801423 0DA09CA4 42AD4FFE 05BFA1BB B1F0A91D  
D306B130  
1D060355 1D0E0416 0414230D A09CA442 AD4FFE05 BFA1BBB1 F0A91DD3  
06B1300D  
06092A86 4886F70D 01010505 00038201 0100B51C 52D0B7F6 3C1C6794  
0CE78169  
1EA24563 77ACC3FB 62E5BD0D 6FC95095 DCB16F18 5E244FB3 34D39712  
86802C28  
B265B47A 73D6ED94 00270982 3D9FD721 C6356EB0 E6C67FC5 0B26017E  
0AA27B11  
B08AD737 4BC2D974 044D2577 9AC759DB 24FC0F9B DD7780F0 70D26A52  
9A3043F6  
31F776F2 1763207E 08B5EDA0 3A541E75 171B37A3 272D90DC D2700D0E  
99D72E07  
007CFB13 93A24BF4 75E3995A 13553515 CAFDACD6 BCB77728 0576A59C  
8B7B0FE1  
CEEAD612 C4D565C3 0184DD1E E8D896C0 2DD8990F 701E71E1 1C63DB6F  
F2BBD4C4  
CC7811C6 D120C631 173FC216 5AEA05E2 D7BA363D B63638F2 476D9F06  
7F59C09B  
B855A632 4C942EC0 05FD59F9 D1CD5AC9 1FBE

quit

license udi pid ISR4321/K9 sn FD021491LXF  
no license smart enable  
diagnostic bootup level minimal  
spanning-tree extend system-id  
redundancy  
mode none  
interface GigabitEthernet0/0/0  
ip address 192.168.0.1 255.255.255.0  
negotiation auto  
ipv6 address 2001:DB8:ACAD::1/64  
ipv6 ospf 1 area 0  
interface GigabitEthernet0/0/1  
ip address 192.168.2.1 255.255.255.0  
negotiation auto  
ipv6 address 2001:DB8:ACAD:2::1/64  
ipv6 ospf 1 area 2  
interface Serial0/1/0  
no ip address  
shutdown

```
interface Serial0/1/1
  no ip address
  shutdown
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto
router ospf 1
  router-id 1.1.1.1
  network 192.168.0.0 0.0.0.255 area 0
  network 192.168.2.0 0.0.0.255 area 2
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
  router-id 1.1.1.1
control-plane
line con 0
  transport input none
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
End
```

```
R2:
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core
hostname R2
boot-start-marker
boot system bootflash:isr4300-universalk9.16.09.08.SPA.bin
boot system flash isr4300-universalk9.16.09.08.SPA.bin
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
  exit-address-family
  address-family ipv6
  exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
```

```
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM240607Q1
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
  mode none
interface GigabitEthernet0/0/0
  ip address 192.168.0.2 255.255.255.0
  negotiation auto
  ipv6 address FE80::1 link-local
  ipv6 address 2001:DB8:ACAD::2/64
  ipv6 ospf 1 area 0
interface GigabitEthernet0/0/1
  ip address 192.168.3.1 255.255.255.0
  negotiation auto
  ipv6 address FE80::1 link-local
  ipv6 address 2001:DB8:ACAD:3::1/64
  ipv6 ospf 1 area 0
interface GigabitEthernet0/1/0
  no ip address
  shutdown
  negotiation auto
interface GigabitEthernet0/1/1
  no ip address
  shutdown
  negotiation auto
interface GigabitEthernet0
  vrf forwarding Mgmt-intf
  no ip address
  shutdown
  negotiation auto
router ospf 1
  router-id 1.1.1.2
  network 192.168.0.0 0.0.0.255 area 0
  network 192.168.3.0 0.0.0.255 area 0
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
control-plane
line con 0
  transport input none
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
End
```

```
R3:
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R3
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
  exit-address-family
  address-family ipv6
  exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-2667303412
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-2667303412
  revocation-check none
  rsakeypair TP-self-signed-2667303412
crypto pki certificate chain TP-self-signed-2667303412
  certificate self-signed 01
    30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
    31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
    69666963 6174652D 32363637 33303334 3132301E 170D3235 30393139
32323033
    30385A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
    4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32
36363733
    30333431 32308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
    0A028201 01009EEC CC803FC8 7566882D D99F20FD 37A32DB6 EE0D115F
096C29C0
    3B145DFB BA9E1E03 B3C19E66 5FD0B14B 21E87D01 633F1149 213CC4B8
24FBF3F2
    223A616B 757B7CDB A82DA6F7 9122A8A5 E726DFA4 E78D4C34 9A5138DB
1D76E5FF
    ACAE3F7C CDC11055 F4298A45 6C772164 F7735990 357FF4F9 1C075414
7EF41B9B
    10954CB8 A9829224 29F50C8F D7C07103 ABF697C9 2A9062FC A960D44E
FE2CBE3D
```



896ED7FF F7E03BA9 0C1E93FC 3BCA4D1B 7A13E524 48C43AA3 85B2ADED  
8E920325  
7156E608 83290214 7EEF90AC F1ECB161 EEFBBD2FB 643FCA11 DAE450B8  
A7FDD3A1  
8976CDEB 472B741F 601EA8E6 1C8AD8B6 D0D411DC AD62B99F F8C2F300  
29745E6D  
DE5AB9B3 E28F0203 010001A3 53305130 0F060355 1D130101 FF040530  
030101FF  
301F0603 551D2304 18301680 14F49928 13F55300 4BB3BB26 E32F1D84  
A309E3B8  
D7301D06 03551D0E 04160414 F4992813 F553004B B3BB26E3 2F1D84A3  
09E3B8D7  
300D0609 2A864886 F70D0101 05050003 82010100 8C688DF7 F55DFC33  
E7D7B17A  
F11A65AC C940E9E6 FF8D5A3E FB966589 DC3CF657 7993255F 0D207D1F  
425A0D59  
CFEB41A1 D69C843C 9D5DE511 24BD79AD 0A60FA2A 16842FC4 922A2DA9  
5FDC4B05  
8629525F 2A615E40 7B4A9C58 C88F5677 3A65971E A06F9926 38CEA7B1  
E4FA87ED  
453532E4 3EC3E5B2 8C046B62 F5234ED5 96AA9160 17AE42D9 8A3E74B5  
6DCC98E0  
8D115A07 34E9B5F2 F9261586 574F4BF8 5C686423 CC40CFBA 84CD7C12  
D3AD4FE8  
110ADA71 F9F31247 BAF3C67E C385CFB2 81741837 EBFDD1C85 B9E8C063  
7B5902E7  
B498EFF4 849525B1 8E068394 D1D53687 94A2E880 9048F595 F094852E  
A91272F0  
76D99CF1 9ACCC66A 065CF9D7 6C5459C7 BFF237E1

quit

license udi pid ISR4321/K9 sn FLM2407011F  
no license smart enable  
diagnostic bootup level minimal  
spanning-tree extend system-id  
redundancy  
mode none  
interface GigabitEthernet0/0/0  
ip address 192.168.3.2 255.255.255.0  
negotiation auto  
ipv6 address 2001:DB8:ACAD:3::2/64  
ipv6 ospf 1 area 0  
interface GigabitEthernet0/0/1  
ip address 192.168.1.3 255.255.255.0  
negotiation auto  
ipv6 address 2001:DB8:ACAD:1::3/64  
ipv6 ospf 1 area 1  
interface GigabitEthernet0/1/0  
no ip address  
shutdown  
negotiation auto  
interface GigabitEthernet0/1/1

```
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 1
 router-id 1.1.1.3
 network 192.168.1.0 0.0.0.255 area 1
 network 192.168.3.0 0.0.0.255 area 0
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login
End
```

```
R4:
hostname R4
boot-start-marker
boot-end-marker
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-2949602955
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-2949602955
revocation-check none
rsa-keypair TP-self-signed-2949602955
crypto pki certificate chain TP-self-signed-2949602955
certificate self-signed 01
 30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
```

05050030  
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D  
43657274  
69666963 6174652D 32393439 36303239 3535301E 170D3235 30393139  
32323334  
33345A17 0D333030 31303130 30303030 305A3031 312F302D 06035504  
03132649  
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32  
39343936  
30323935 35308201 22300D06 092A8648 86F70D01 01010500 0382010F  
00308201  
0A028201 0100BEB0 6A33EBB9 1DAA379F D1E25BF8 BC59C100 AAF5CA43  
59B1B2AB  
F6D419CB 082B7D0F 1567B86D 06077EFD B86916CA C1FA6C8B FE37DDE0  
5D0BE541  
8E311A5A 3F7BEBE1 213555C4 B492A4B0 96AAAE94 DD96A9AC C45A7267  
71FBC162  
5F4D5982 95BF3111 B0EEAAB5 B8D91A5D D48E00C8 5F45B83E E48746CD  
77B0277C  
C051BCAA EDE02DCF C93B3596 CE8D3E69 47773787 84C4DD22 8BADA484  
260C01DC  
EC8DB955 244BA9F6 D934ADEA 7B9ACA21 6F651B50 CB462EB1 EF9977F7  
9F69C2A1  
FB444FE8 FC8FABF4 CE81AAF2 EB3F458E ECF95C20 3F18754F C3E51D91  
ADBD FE63  
7687BEBB 2FDD454E 964190C5 D3BFE1AF 9F9E93BF 1CE9B5CC F9DBF98F  
9B209CD1  
4D2BDAED EFED0203 010001A3 53305130 0F060355 1D130101 FF040530  
030101FF  
301F0603 551D2304 18301680 1416778F F554F142 9D50D942 7666C5F9  
38CCEE25  
E1301D06 03551D0E 04160414 16778FF5 54F1429D 50D94276 66C5F938  
CCEE25E1  
300D0609 2A864886 F70D0101 05050003 82010100 8080264D 1DC53EAA  
BB907A8F  
4F5885F6 D2CA8F65 DAB6BB4A 278F4D96 00F16AED 6CDF2CB0 23BAEF09  
FC14F3C5  
F8CECC73 DF731A56 D6052A76 F563F880 376A391A 60FC9B65 70747D0B  
C24BC0E8  
C09E66B4 5074B19B 01A43F16 43866452 2A117CA5 F5CA027F CE5AE4AC  
C1ADC1B8  
EE475640 0980C2BA 9DDFD59A 0D50E70E FD160129 A4E561B8 EB6FAE10  
64FDB11C  
C733684F A6941782 E96B7D8B CF5FF4D9 387D0C6F 37A023E9 687988D9  
ED564FDD  
25DF8226 90261F26 0019C0C8 57CBFBA4 4E5699D6 2B619549 DE127B3C  
E45DAC5A  
3DFD297B EACED812 30B7FC5F 26A6D4B1 A280C6B2 24142203 DC953289  
A1E46AF1  
C0A33C81 26E22A76 74C2DDAC BAB28330 988D6262  
quit

```
license udi pid ISR4321/K9 sn FD0214420HW
license boot level appxk9
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.1.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:1::2/64
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
ip address 192.168.5.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:5::1/64
ipv6 ospf 1 area 1
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
router ospf 1
router-id 1.1.1.4
network 192.168.1.0 0.0.0.255 area 1
network 192.168.5.0 0.0.0.255 area 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
router-id 1.1.1.4
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

```
R5:
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
```

```
hostname R5
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
  address-family ipv4
  exit-address-family
  address-family ipv6
  exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-214699181
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-214699181
  revocation-check none
  rsakeypair TP-self-signed-214699181
crypto pki certificate chain TP-self-signed-214699181
  certificate self-signed 01
    3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
    30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
    69666963 6174652D 32313436 39393138 31301E17 0D323530 39313932
33333432
    335A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403
1325494F
    532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3231
34363939
    31383130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02
    82010100 8802AFE7 200B353B B28074ED E9E103C7 BDACAEED 7AF6363A
6A6EAB62
    DF06BC85 CEFB9B8B 94B95109 5EB3EF92 30F1C24F 4CCE5733 21EF2763
BFA08A81
    8704FEC7 56475291 33AEDE38 FA443A7A 549149F8 F124BD1C 8CEE5062
1B566FA1
    F9D5771C CEA516EE AC03C7A4 9F385B7B 0E7C430B 78955379 512646B9
10FACB38
    685556F3 D9AF7EE0 3E056CF7 2293873C 584E24BB 249F937B 7FF5410B
B1495644
    14EAE3B9 39631F66 9796D994 7DDFABCC 9F173D3C 46A34894 099B139D
906F3210
    800A450E B420E8C1 B8CE3B56 E41F2BEC 851A6E78 16ECB492 5EFF4366
395A1C2B
    844FDF3C 85020CF5 08A1C6F8 6FFBB76F 08CD4D30 93F27850 698B33E4
A83DDA31
```

```
AE068A3F 02030100 01A35330 51300F06 03551D13 0101FF04 05300301
01FF301F
0603551D 23041830 1680141F B40A624D C9C97665 660BDE2A C260CF09
0097D130
1D060355 1D0E0416 04141FB4 0A624DC9 C9766566 0BDE2AC2 60CF0900
97D1300D
06092A86 4886F70D 01010505 00038201 01003616 3CB1C1A7 BC88BD3B
466DA917
428F9E65 F4B5D6CB BDD11871 7C90E80E F8731B0A AC943B65 3E656BA2
6606B21F
E1F285AE C0D5BEF6 85E933A8 3F2F2DD0 AA6FAEC1 1651CB9A AAF43F04
4A3DDE93
FC91C218 BC4F0D9F 5D8A9058 80DB650C 8B13E1EA FE67155D 8D604C5A
AA56D32F
45948321 C14A8380 B9768A52 58B26520 867C783F D4C7B584 924C6BA8
69278AD1
3FBD2CC4 EBA7FDA1 49A778C9 CC64E026 0D024EEB 4ECAA4E3 D0A03B31
45A3D1CD
449C9EDB 9D8E9F1D A2A55A04 AD96F6C5 BF4953F1 F2D98957 1EAA4FBE
B230CDAE
C8081876 111C6B1F 7D88A71D 3FE1FF67 5FCEB989 53815457 721EBC39
639413C3
50FBDCEB 81E22F4D 9D3E46EB 078DB219 E4FF
quit
license udi pid ISR4321/K9 sn FD021442167
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.5.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:5::2/64
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
no ip address
negotiation auto
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 1
router-id 1.1.1.5
```

```
network 192.168.5.0 0.0.0.255 area 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
control-plane
line con 0
  transport input none
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login
end
```

```
MLS 1:
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
hostname MLS1
boot-start-marker
boot-end-marker
no aaa new-model
system mtu routing 1500
ip routing
ipv6 unicast-routing
vtp domain CCNP
vtp mode transparent
spanning-tree mode pvst
spanning-tree extend system-id
interface FastEthernet1/0/1
  no switchport
  ip address 192.168.12.2 255.255.255.0
  ipv6 address 2001:DB8:ACAD:2::2/64
  ipv6 ospf 1 area 2
interface FastEthernet1/0/2
  no switchport
  ip address 192.168.2.2 255.255.255.0
  ipv6 address 2001:DB8:ACAD:12::2/64
  ipv6 ospf 1 area 2
interface FastEthernet1/0/3
interface FastEthernet1/0/4
interface FastEthernet1/0/5
interface FastEthernet1/0/6
interface FastEthernet1/0/7
interface FastEthernet1/0/8
interface FastEthernet1/0/9
```

```
interface FastEthernet1/0/10
interface FastEthernet1/0/11
interface FastEthernet1/0/12
interface FastEthernet1/0/13
interface FastEthernet1/0/14
interface FastEthernet1/0/15
interface FastEthernet1/0/16
interface FastEthernet1/0/17
interface FastEthernet1/0/18
interface FastEthernet1/0/19
interface FastEthernet1/0/20
interface FastEthernet1/0/21
interface FastEthernet1/0/22
interface FastEthernet1/0/23
interface FastEthernet1/0/24
interface GigabitEthernet1/0/1
interface GigabitEthernet1/0/2
interface GigabitEthernet1/1/1
    speed auto 1000
interface GigabitEthernet1/1/2
interface Vlan1
    no ip address
router ospf 1
    router-id 1.1.1.7
    network 192.168.2.0 0.0.0.255 area 2
    network 192.168.12.0 0.0.0.255 area 2
ip http server
ip http secure-server
logging esm config
ipv6 router ospf 1
    router-id 1.2.1.1
line con 0
line vty 5 15
End
```

```
Mls 2 :
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
hostname S2
boot-start-marker
boot-end-marker
no aaa new-model
system mtu routing 1500
ip routing
ipv6 unicast-routing
vtp domain CCNP
vtp mode transparent
mpls label protocol ldp
```



```
crypto pki trustpoint TP-self-signed-3180753792
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-3180753792
  revocation-check none
  rsakeypair TP-self-signed-3180753792
crypto pki certificate chain TP-self-signed-3180753792
  certificate self-signed 01
    3082023E 308201A7 A0030201 02020101 300D0609 2A864886 F70D0101
04050030
    31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
    69666963 6174652D 33313830 37353337 3932301E 170D3933 30333031
30303031
    30315A17 0D323030 31303130 30303030 305A3031 312F302D 06035504
03132649
    4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D33
31383037
    35333739 3230819F 300D0609 2A864886 F70D0101 01050003 818D0030
81890281
    8100A334 3E4304F6 BAB68601 D9A55BA4 0898A0D3 8D6A82A4 65BE961E
6C85AC95
    2DC4532A 706862C1 D1C659E0 B901D856 80C9E308 65D10954 BE74FB7E
4A226029
    415336FF C1426961 13DE2D5D 5D1DC511 17283CE7 BAC499D7 2763EC77
280A11F3
    B01EEC85 1192D3C3 2 03B18F9B D6B9231A E62F8B30 E1574139 BCB6A764
C19696E2 D77DA23D BB292DF2C709C5B 8F373BD1 9DE67197 4492C07B 1C21DDC4
3BFFB73D
    C00B0203 010001A3 66306430 0F060355 1D130101 FF040530 030101FF
30110603
    551D1104 0A300882 06537769 74636830 1F060355 1D230418 30168014
62DE0870
    F1B30250 A1F39293 9EFB7884 1179817B 301D0603 551D0E04 16041462
DE0870F1
    B30250A1 F392939E FB788411 79817B30 0D06092A 864886F7 0D010104
05000381
    810010C7 20FD0414 C0BA58C0 190F2C6C CC476316 80C1A6DC E917FBF5
13982184
    427732C1 B0666AB6 C018A9CA 27B6B042 C2D078EF 106F2C3F 25807BB1
C0A92844
    C8AE2A1E 202538B0 B05B89B5 99AA853D A7F137C2 A7F23A7A 2426595E
78E1D8F2
    B50976DB 3BCFD52E 6E53F865 41EC3EF8 9A6AC0CD 7CC59419 62014FD0 1596C00D
0E5E
quit
spanning-tree mode pvst
spanning-tree extend system-id
interface FastEthernet1/0/1
  no switchport
  ip address 192.168.12.3 255.255.255.0
  ipv6 address 2001:DB8:ACAD:6::3/64
```

```
ipv6 ospf 1 area 2
interface FastEthernet1/0/2
interface FastEthernet1/0/3
interface FastEthernet1/0/4
interface FastEthernet1/0/5
interface FastEthernet1/0/6
interface FastEthernet1/0/7
interface FastEthernet1/0/8
interface FastEthernet1/0/9
interface FastEthernet1/0/10
interface FastEthernet1/0/11
interface FastEthernet1/0/12
interface FastEthernet1/0/13
interface FastEthernet1/0/14
interface FastEthernet1/0/15
interface FastEthernet1/0/16
interface FastEthernet1/0/17
interface FastEthernet1/0/18
interface FastEthernet1/0/19
interface FastEthernet1/0/20
interface FastEthernet1/0/21
interface FastEthernet1/0/22
interface FastEthernet1/0/23
interface FastEthernet1/0/24
interface GigabitEthernet1/0/1
interface GigabitEthernet1/0/2
interface GigabitEthernet1/1/1
    speed auto 1000
interface GigabitEthernet1/1/2
    no switchport
    ip address 10.10.3.1 255.255.255.0
    mpls label protocol ldp
    mpls ip
interface Vlan1
    no ip address
router ospf 1
    router-id 1.1.2.1
    network 192.168.12.0 0.0.0.255 area 2
ip http server
ip http secure-server
logging esm config
ipv6 router ospf 1
    router-id 1.1.2.1
line con 0
line vty 0 4
    login
line vty 5 15
    login
end
```

## **Problems:**

### **Misconfigured networks**

This was my first networking lab in about 3-4 months. Consequently I had forgotten that each individual port on a router should represent a different network, and after much trial and error, I was able to figure out how to put a different network on every port.

### **Lack of knowledge on the working of multi area ospf and ospf**

Going into this lab, I had forgotten a lot about how to configure ospf and multi area ospf. It took a few hours of research in order to figure out the correct configurations for multi area ospf for both ipv4 and ipv6.

### **Faulty routers**

The rack of routers that I was working on had several faulty routers, it took a several days to troubleshoot and find out how to work around the faulty devices.

### **IPv4 and IPv6**

We typed in multiple incorrect ip addresses. This is the issue that likely took us the longest to fix, as it took us close to a week to find out which of the couple dozen ip addresses had been configured. As this took us so long to debug, I will now be writing down all the ip addresses that I will use in my network configurations.

### **Shared workspace**

Because the rack we were using was shared between multiple different groups, we had to wait hours reloading and reconfiguring the routers with our configurations.